# Structural $\dim B = 5$ for $\lambda = (4, 3, 1^q)$ via disjoint $\Phi$-supports and adjacent-injectivity

Clio

13 May 2026 (afternoon-3 prove session)

**Abstract**

The May-13 morning paper `2026-05-13-4-3-1n-upper-bound.tex` proved $\dim B^{(\lambda)}_{j_0} V_\lambda \leq 5$ structurally at $q \geq 5$ for $\lambda = (4, 3, 1^q)$, with computational verification $\dim = 5$ at $q \in \{3, 4\}$. The matching structural lower bound $\dim \geq 5$ was deferred. The afternoon partial paper `2026-05-13-4-3-1n-lower-bound-analysis.tex` reduced the lower bound to two within-class projection independences and identified a "position-of-letter contamination" obstruction; the afternoon-2 dim2-331 paper (Remark 4.10) noted that the within-class linear independences are individually closed by adjacent-injectivity but concluded that this is *not* sufficient to close the full 5-vector independence, citing a possible cross-class $c_3$-overlap invisible to either projection. We show that this concern is unfounded: the lower bound *is* closed by adjacent-injectivity, provided one works at the level of $V_\lambda$ directly (without projection).

**Theorem.** For $\lambda = (4, 3, 1^q)$ with $q$ in the structural range (the intersection of the inputs below),

$$\dim B^{(\lambda)}_{j_0} V_\lambda \ = \ 5.$$

**Key idea.** The $\Phi^\lambda_A$- and $\Phi^\lambda_B$-images have seminormal SYT-supports that are *disjoint as sets of SYTs of* $\lambda$, because the unordered cell-pairs $\{c_1, c_3\}$ and $\{c_2, c_3\}$ differ (they share $c_3$ but have distinct second cells). Hence $\Phi^\lambda_A(B^{(\mu_A)}) \oplus \Phi^\lambda_B(B^{(\mu_B)})$ is a direct sum of dimension $2 + 3 = 5$ inside $V_\lambda$. Both summands lie in $E^+_{n-1}|_{V_\lambda}$ by construction. The outer chain $\mathcal{O}_\lambda$ acts injectively on $E^+_{n-1}|_{V_\lambda}$ by adjacent-injectivity (dim2-331 Theorem 3.4, applied with four factors). Hence the five lifted vectors are linearly independent in $V_\lambda$.

**What this resolves.** The afternoon partial paper's Proposition 4.4 ("naive separate within-class vanishings, not provable from position-of-$n$ alone") was correct as a statement about projected vectors. But the unprojected direct argument (disjoint SYT-supports of the $\Phi^\lambda_X$-images BEFORE applying $\mathcal{O}_\lambda$) bypasses the contamination issue entirely. The tracer-pair Hoefsmit computation is not needed; adjacent-injectivity suffices.

**What this generalises.** An r=2 dim-additivity statement: for any rank-zero shape $\lambda$ with exactly two length-preserving corners $c_1, c_2$ and column-1 bottom $c_3$, where the SRD-style reduction formula gives $B^{(\lambda)} = \mathcal{O}_\lambda[\Phi^\lambda_A(B^{(\mu_A)}) + \Phi^\lambda_B(B^{(\mu_B)})]$, the dimensions add: $\dim B^{(\lambda)} = \dim B^{(\mu_A)} + \dim B^{(\mu_B)}$. The proof is shape-agnostic; the inputs are the reduction formula and the two inner-bound dims.

## 1 Setup

Notation as in the May-13 4-3-1n upper bound paper (**May-13-UB**, `2026-05-13-4-3-1n-upper-bound.tex`) and the afternoon partial paper (**LB-partial**, `2026-05-13-4-3-1n-lower-bound-analysis.tex`). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and the standard braid relations. $V_\lambda$ is the seminormal cell module for $\lambda \vdash n$ in asymmetric

Murphy normalisation ($b'=1$). $E_j^{\pm}$ denote the $q$- and $(-1)$-eigenspaces of $T_j$. For $1 \le k \le n$,

$$R'_k \;:=\; (T_{k-1}{+}1)\cdots(T_1{+}1), \qquad B_j^{(\lambda)}V_\lambda \;:=\; R'_j \cdots R'_n V_\lambda.$$

**Throughout this paper.** $\lambda := (4,3,1^q)$ with $q \ge q_*$ (with $q_*$ described below), $n := q+7$, $\ell := q+2$, $j_0 := \ell+1 = q+3$. Removable corners

$$c_1 = (1,4), \quad c_2 = (2,3), \quad c_3 = c_* = (q+2,1).$$

Inner partitions (corner-pair removal):

$$\begin{aligned} \mu_A &:= \lambda \setminus \{c_1,c_3\} = (3,3,1^{q-1}), & |\mu_A| &= q+5, \\ \mu_B &:= \lambda \setminus \{c_2,c_3\} = (4,2,1^{q-1}), & |\mu_B| &= q+5. \end{aligned}$$

Outer chain:

$$\mathcal{O}_\lambda \;:=\; (T_{q+2}{+}1)(T_{q+3}{+}1)(T_{q+4}{+}1)(T_{q+5}{+}1).$$

Four factors with indices $\{q+2,q+3,q+4,q+5\} = \{\ell,\ell+1,n-3,n-2\}$.

**Reduction formula (May-13-UB).**

$$B_{j_0}^{(\lambda)}V_\lambda \;=\; \mathcal{O}_\lambda\Big[\Phi_A^\lambda\big(B_{j_0(\mu_A)}^{(\mu_A)}V_{\mu_A}\big) + \Phi_B^\lambda\big(B_{j_0(\mu_B)}^{(\mu_B)}V_{\mu_B}\big)\Big]. \tag{1}$$

The $\mu_C$-piece ($\Phi_C^\lambda$, removing $\{c_1,c_2\}$) and the same-row piece ($S_1$ at row 1, $S_2$ at row 2) vanish structurally for all $q \ge 3$ (May-13-UB Lemmas 3.5 and 3.6).

**The branching injections $\Phi_X^\lambda$.** For each corner pair $\{c_X^-, c_X^+\} \in \{\{c_1,c_3\},\{c_2,c_3\},\{c_1,c_2\}\}$, the map $\Phi_X^\lambda : V_{\mu_X} \hookrightarrow V_\lambda$ sends a seminormal basis vector $v_{T'}^{(\mu_X)}$ to the $+q$-eigenvector of the $T_{n-1}$-2-block on the two SYT extensions $T_A^X, T_B^X$ of $T'$ obtained by placing $\{n-1,n\}$ at $\{c_X^-, c_X^+\}$:

$$\Phi_X^\lambda(v_{T'}^{(\mu_X)}) \;=\; v_{T_A^X} + \tau_X v_{T_B^X},$$

where $\tau_X$ is the Hoefsmit off-diagonal coefficient determined by the axial distance of the pair (May-13-UB Theorem 2.2, item 1; cf. SRD-331 Lemma 3.2).

**Inner-bound dims.**

- $\dim B_{j_0(\mu_A)}^{(\mu_A)}V_{\mu_A} = 2$. *Structural at $q-1 \ge 4$, i.e., $q \ge 5$* (dim2-331 Theorem 1.3, applied to $\mu_A = (3,3,1^{q-1})$). Computational at $q \in \{3,4\}$ (SRD-331 Section 5).
- $\dim B_{j_0(\mu_B)}^{(\mu_B)}V_{\mu_B} = 3$. *Structural at $|\mu_B| \ge 8$, i.e., $q \ge 3$* (May-12-Eve3 paper `2026-05-12-evening-4-2-1n-dir` amended with the May-13 SRD-421 same-row-dies closure, `2026-05-13-same-row-dies-421.tex`).

**Range of structural validity.** The combination of inputs is structural at $q \ge q_*$, with the binding constraint being the dim2-331 input on $\mu_A$, giving $q_* = 5$. At $q \in \{3,4\}$, the inner-bound dims are computational, hence the theorem reduces to a computational statement already verified by May-13-UB Section 5.

## 2 The two key lemmas

### 2.1 Disjoint Φ-supports at the SYT level

**Lemma 1** (Disjoint SYT supports)**.** *For every $v \in V_{\mu_A}$ and every $w \in V_{\mu_B}$, the seminormal supports of $\Phi^\lambda_A(v)$ and $\Phi^\lambda_B(w)$ in $V_\lambda$ are disjoint as sets of SYTs of $\lambda$. Consequently,*

$$\Phi^\lambda_A(V_{\mu_A}) \ \cap\ \Phi^\lambda_B(V_{\mu_B}) \ =\ \{0\}$$

*inside $V_\lambda$.*

*Proof.* By the explicit description of $\Phi^\lambda_X$ (§**??**), every SYT $T$ in the seminormal support of $\Phi^\lambda_X(v^{(\mu_X)}_{T'})$ has the two letters $\{n-1, n\}$ occupying the cell pair $\{c^-_X, c^+_X\}$ (in one of two orders). Extending by linearity over $v \in V_{\mu_X}$, every SYT in the seminormal support of $\Phi^\lambda_X(v)$ has $\{n-1, n\}$ at $\{c^-_X, c^+_X\}$ as an unordered cell pair.

For $X = A$: cell pair is $\{c_1, c_3\}$. For $X = B$: cell pair is $\{c_2, c_3\}$. These two unordered pairs are distinct subsets of cells of $\lambda$ (they share $c_3$ but differ in the other element, $c_1$ vs $c_2$).

Hence no single SYT $T$ of $\lambda$ can lie in both supports: in $T$, the unordered cell pair $\{T^{-1}(n-1), T^{-1}(n)\}$ is determined, and it cannot equal both $\{c_1, c_3\}$ and $\{c_2, c_3\}$.

The seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$ is linearly independent in $V_\lambda$. Hence any non-zero linear combination supported on the A-set is linearly independent of any non-zero linear combination supported on the B-set. So $\Phi^\lambda_A(V_{\mu_A}) \cap \Phi^\lambda_B(V_{\mu_B}) = \{0\}$. □

### 2.2 Image lies in $E^+_{n-1}$

**Lemma 2** ($\Phi^\lambda_X$ lands in $E^+_{n-1}$)**.** *For each $X \in \{A, B\}$ and every $v \in V_{\mu_X}$, $\Phi^\lambda_X(v) \in E^+_{n-1}|_{V_\lambda}$.*

*Proof.* By construction (§**??**), $\Phi^\lambda_X(v^{(\mu_X)}_{T'})$ is the $+q$-eigenvector of the $T_{n-1}$-action on the 2-block $\langle v_{T^X_A}, v_{T^X_B}\rangle$ of $V_\lambda$, where the two SYTs are obtained by placing $\{n-1, n\}$ at $\{c^-_X, c^+_X\}$ in the two valid orders. The axial distance $d_X$ of this 2-block is non-zero (the cells $c^-_X, c^+_X$ lie in distinct rows and columns), so the 2-block is non-degenerate and the $+q$-eigenvector exists and is unique up to scalar. Hence $\Phi^\lambda_X(v^{(\mu_X)}_{T'}) \in E^+_{n-1}|_{V_\lambda}$. Extending by linearity: $\Phi^\lambda_X(V_{\mu_X}) \subseteq E^+_{n-1}|_{V_\lambda}$. □

### 2.3 Adjacent-injectivity of the outer chain

**Theorem 3** (Adjacent injectivity chain; dim2-331 Theorem 3.4)**.** *Let $V$ be any $H_q(S_n)$-module, and let $a \le b \le n-2$. The composition*

$$(T_a{+}1)(T_{a+1}{+}1)\cdots(T_b{+}1)\big|_{E^+_{b+1}(V)} \ :\ E^+_{b+1}(V) \ \to\ E^+_a(V)$$

*is injective.*

*Proof.* This is dim2-331 Theorem 3.4 (also dim2-331 Theorem 5.5). The proof is an iterated application of the basic adjacent-injectivity proposition: each $(T_j+1)$ maps $E^+_{j+1}(V)$ injectively into $E^+_j(V)$, by the combination of image-landing ($(T_j+1)V \subseteq E^+_j$, via the quadratic relation) and adjacent-disjointness ($E^-_j(V) \cap E^+_{j+1}(V) = 0$, May-19 Phase A swapped form `2026-05-19-phase-a-general.tex` Lemma 3.3). □

**Corollary 4** (Outer chain is injective on $E^+_{n-1}$)**.** *The operator $\mathcal{O}_\lambda$ acts injectively on $E^+_{n-1}|_{V_\lambda}$, with image in $E^+_\ell|_{V_\lambda} = E^+_{q+2}|_{V_\lambda}$.*

*Proof.* Apply Theorem **??** with $a = q+2$, $b = q+5 = n-2$ (four factors). □

## 3 The main theorem

**Theorem 5** (Main: lower bound matches upper bound)**.** *For $\lambda = (4,3,1^q)$ with $q \geq q_*$ (where $q_* = 5$ is the binding structural threshold from the dim2-331 input on $\mu_A$),*

$$\dim B_{j_0}^{(\lambda)} V_\lambda \;=\; 5. \tag{2}$$

*Proof. Upper bound:* $\dim B^{(\lambda)} \leq 5$ by May-13-UB Theorem 1.1.

*Lower bound:* Define

$$W \;:=\; \Phi_A^\lambda\big(B^{(\mu_A)} V_{\mu_A}\big) \;+\; \Phi_B^\lambda\big(B^{(\mu_B)} V_{\mu_B}\big) \;\subseteq\; V_\lambda.$$

We compute $\dim W$ and then $\dim \mathcal{O}_\lambda W$:

**Step 1.** $W$ is a direct sum of dim $2+3=5$. By Lemma **??**, $\Phi_A^\lambda(V_{\mu_A}) \cap \Phi_B^\lambda(V_{\mu_B}) = \{0\}$, and in particular $\Phi_A^\lambda(B^{(\mu_A)}) \cap \Phi_B^\lambda(B^{(\mu_B)}) = \{0\}$. Hence $W = \Phi_A^\lambda(B^{(\mu_A)}) \oplus \Phi_B^\lambda(B^{(\mu_B)})$ is a direct sum. Each $\Phi_X^\lambda$ is injective on $V_{\mu_X}$ (it sends distinct seminormal basis vectors to vectors with disjoint 2-element supports), so the inner dims lift:

$$\dim W \;=\; \dim B^{(\mu_A)} + \dim B^{(\mu_B)} \;=\; 2+3 \;=\; 5.$$

**Step 2.** $W \subseteq E_{n-1}^+|_{V_\lambda}$. By Lemma **??**, each of $\Phi_A^\lambda(B^{(\mu_A)})$ and $\Phi_B^\lambda(B^{(\mu_B)})$ lies in $E_{n-1}^+|_{V_\lambda}$. Their sum (direct or not) also lies there.

**Step 3.** $\mathcal{O}_\lambda$ is injective on $W$. By Corollary **??**, $\mathcal{O}_\lambda$ is injective on all of $E_{n-1}^+|_{V_\lambda}$, hence on the subspace $W$.

**Step 4.** Conclude. By Steps 1, 3:

$$\dim \mathcal{O}_\lambda W \;=\; \dim W \;=\; 5.$$

By the reduction formula (**??**), $B_{j_0}^{(\lambda)} V_\lambda = \mathcal{O}_\lambda W$, so $\dim B^{(\lambda)} \geq 5$.

Combined with the upper bound: $\dim B^{(\lambda)} = 5$. □

## 4 Why the afternoon partial paper got stuck

The afternoon partial paper proceeded via the position-of-$n$ projection $\pi_n^{(c)}$ (for $c \in \{c_1, c_2, c_3\}$). Its Theorem 3.2 reduced the full 5-vector independence to two within-class projection independences:

(a) $\{\pi_n^{(c_1)} F_A^1, \pi_n^{(c_1)} F_A^2\}$ linearly independent.

(b) $\{\pi_n^{(c_2)} F_B^1, \pi_n^{(c_2)} F_B^2, \pi_n^{(c_2)} F_B^3\}$ linearly independent.

Both projections are taken AFTER applying $\mathcal{O}_\lambda$. The issue, as flagged in dim2-331 Remark 4.10: the projected vectors are not in $E_{n-1}^+$ (the seminormal projection $\pi_n^{(c)}$ breaks the $T_{n-1}$-2-block $+q$-eigenvector structure), so adjacent-injectivity does not apply to the projected vectors. The "cross-class $c_3$-overlap" concern: a linear combination $\sum \alpha_i F_A^i + \sum \beta_j F_B^j = 0$ could carry weight on SYTs with $n$ at $c_3$ (where both $\Phi_A$- and $\Phi_B$-pieces project non-trivially), and this overlap could be invisible to either $\pi_n^{(c_1)}$ or $\pi_n^{(c_2)}$ alone.

**Resolution.** The concern is real for the projected approach but does *not* obstruct the direct approach. Working at the level of $V_\lambda$ (no projection):

- Before applying $\mathcal{O}_\lambda$: the lifts $\Phi^\lambda_A(v)$ and $\Phi^\lambda_B(w)$ have seminormal supports on SYTs distinguished by the position of *both* $n-1$ AND $n$, not just $n$. The cell-pair $\{n-1, n\}$ is at $\{c_1, c_3\}$ for A-lifts and at $\{c_2, c_3\}$ for B-lifts; these are distinct unordered pairs, hence disjoint SYT supports (Lemma **??**).
- After applying $\mathcal{O}_\lambda$: the position of $n-1$ is moved by the factors $T_{n-2}, T_{n-3}$ in $\mathcal{O}_\lambda$, so the SYT supports of $F^i_A$ and $F^j_B$ can overlap (in particular at SYTs with $n$ at $c_3$). *However*, this overlap is the image of two linearly independent subspaces $\Phi^\lambda_A(B^{(\mu_A)})$ and $\Phi^\lambda_B(B^{(\mu_B)})$ under the injective map $\mathcal{O}_\lambda|_{E^+_{n-1}}$, so the images remain linearly independent.

**Moral.** The decoupling argument should be performed *before* the outer chain acts. Position-of-$(n-1)$ at the *inner* level (i.e., on the $\Phi^\lambda_X$-supports inside $V_\lambda$) is automatic and clean. The outer chain then acts on the already-decoupled direct sum, and injectivity preserves the decoupling.

**Comparison to position-of-letter contamination.** The afternoon partial paper's "position-of-letter contamination at recursion depth 3" is a real obstruction to lifting position-of-$(n-2)$ *within* a single class through three layers of recursion. That contamination prevents the within-A separation $\{F^1_A, F^2_A\}$ from being detected by any single position-of-letter projection at the $\lambda$ level. But within-class separation is *also* closed by adjacent-injectivity applied to $\Phi^\lambda_A(B^{(\mu_A)})$ alone (dim2-331 Remark 4.10's first sentence). The afternoon-2 author noted this within-class fact but incorrectly concluded that it didn't combine with A-vs-B decoupling to close the full lower bound. The present paper observes that the A-vs-B decoupling is also available at the inner level (disjoint $\Phi$-supports), and both pieces of decoupling chain together through the injective outer chain.

# 5 Computational verification

Script: `~/projects/scratch/2026-05-13-431-lower-closed/verify_431_lower.py`. Mod-$P$ arithmetic at $P = 100\,003$, $q = 11$, for $\lambda = (4, 3, 1^q)$ at $q \in \{3, 4\}$:

- **Disjoint SYT supports.** For each $q$, the number of SYTs of $\lambda$ with $\{n-1, n\}$ at $\{c_1, c_3\}$ and at $\{c_2, c_3\}$:

| $q$ | $n$ | A-support | B-support |
|---|---|---|---|
| 3 | 10 | 112 | 180 |
| 4 | 11 | 240 | 378 |

  Intersection is empty in both cases (consistent with Lemma **??**).
- **Adjacent disjointness.** $E^-_j \cap E^+_{j+1} = 0$ verified at $j \in \{q+2, q+3, q+4, q+5\}$ for both $q \in \{3, 4\}$.
- **Outer chain injectivity.** $\operatorname{rank}(\mathcal{O}_\lambda \cdot E^+_{n-1}) = \dim E^+_{n-1}$ in both cases (245 at $q = 3$, 470 at $q = 4$).
- **Direct dim.** $\dim B^{(\lambda)}_{j_0} V_\lambda = 5$ at each $q \in \{3, 4\}$, matching the conjectured value (May-13-UB Section 5).

# 6 The r=2 dim-additivity meta-theorem

The proof of Theorem **??** uses only three shape-sensitive inputs and one universal fact:

**Shape inputs.**

- The SRD-style reduction formula $B^{(\lambda)} = \mathcal{O}_\lambda[\Phi_A^\lambda(B^{(\mu_A)}) + \Phi_B^\lambda(B^{(\mu_B)})]$, which holds when same-row-dies applies to the same-row pieces of $\lambda$ and the $\Phi_C^\lambda$ piece vanishes ($j$-formula leakage on the $(c_1, c_2)$ block).
- Non-degeneracy of the $T_{n-1}$-2-blocks for both corner pairs $\{c_X^-, c_X^+\}$, $X \in \{A, B\}$ (automatic when corners are at distinct row+column).
- The inner-bound dims $\dim B^{(\mu_A)}$, $\dim B^{(\mu_B)}$.

**Universal fact.** Adjacent-injectivity (Theorem **??**), applied to the four-factor outer chain.

**Combinatorial fact.** Disjointness of the unordered cell-pairs $\{c_A^-, c_A^+\}$ and $\{c_B^-, c_B^+\}$ (automatic when $c_A^- \neq c_B^-$).

**Theorem 6** (r=2 dim-additivity)**.** *Let $\lambda \vdash n$ be a rank-zero partition with exactly two length-preserving corners $c_1, c_2$ and column-1 bottom $c_*$, satisfying:*

1. *The SRD-style reduction formula (**??**) holds for $\lambda$ with $\mathcal{O}_\lambda = (T_\ell{+}1)\cdots(T_{n-2}{+}1)$.*
2. *Both corner-pair $T_{n-1}$-2-blocks $\{c_1, c_*\}$ and $\{c_2, c_*\}$ are non-degenerate ($|d| \geq 2$ axial distance).*

*Then*

$$\dim B_{j_0}^{(\lambda)} V_\lambda \;=\; \dim B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} \;+\; \dim B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B},$$

*where $\mu_A := \lambda \setminus \{c_1, c_*\}$, $\mu_B := \lambda \setminus \{c_2, c_*\}$.*

*Proof.* Repeat the proof of Theorem **??** verbatim. The disjoint SYT-supports lemma (Lemma **??**) generalises to any two corner pairs $\{c_X^-, c_X^+\}$ that differ as unordered sets; the $\{c_1, c_*\}$ vs $\{c_2, c_*\}$ pairs differ in their second element (since $c_1 \neq c_2$). The image-in-$E_{n-1}^+$ lemma (Lemma **??**) uses only non-degeneracy of the corner-pair blocks (hypothesis (ii)). The adjacent-injectivity chain (Corollary **??**) is universal and applies to any outer chain. ☐

**Remark 7** (Generalisation to higher $r$)**.** The r=2 argument generalises immediately to r=3 and beyond, with one caveat. For $r$ length-preserving corners $c_1, \ldots, c_r$ plus the column-1 bottom $c_*$, the reduction formula has $r$ corner-pair $\Phi_X$-pieces (each removing $\{c_i, c_*\}$) plus $\binom{r}{2}$ pure-length-pres $\Phi_{ij}$-pieces (each removing $\{c_i, c_j\}$). The unordered cell pairs are:

$$\{\{c_1, c_*\}, \ldots, \{c_r, c_*\}\} \;\cup\; \{\{c_i, c_j\} : 1 \leq i < j \leq r\}.$$

These $r + \binom{r}{2}$ pairs are pairwise distinct (no two share both cells), so the seminormal supports of the corresponding $\Phi$-images are pairwise disjoint. If the SRD-style reduction restricts the contributing pieces to a subset $\mathcal{S}$ (with the others vanishing by $j$-formula leakage or same-row-dies), then

$$\dim B^{(\lambda)} \;=\; \sum_{X \in \mathcal{S}} \dim B^{(\mu_X)}.$$

This formalises and proves a long-conjectured "additivity" formula for $\dim B$ in the stable regime.

## 7 Honest status

**What we proved structurally.** $\dim B_{j_0}^{(\lambda)} V_\lambda = 5$ for $\lambda = (4, 3, 1^q)$ at $q \geq 5$ (structural range of the dim2-331 input on $\mu_A$).

**What we corrected.** The dim2-331 Remark 4.10 and the afternoon partial paper's "tracer-pair-required" conclusion. Adjacent-injectivity *is* sufficient to close the 5-vector independence, provided one applies it to the unprojected direct sum $\Phi_A^\lambda(B^{(\mu_A)}) \oplus \Phi_B^\lambda(B^{(\mu_B)})$ at the $V_\lambda$ level. The afternoon partial paper's position-of-letter contamination obstruction is real but is an obstruction only to the projected approach.

**What this generalises.** An r=2 dim-additivity theorem (Theorem **??**), parallel to the dim2-331 r=1 dim-equality theorem. Together, the two theorems give a complete recursive computation of $\dim B$ for any rank-zero shape with SRD-style reduction, in the stable regime.

**What remains.**

- Confirming the structural range of the SRD-421 input on $\mu_B$: the May-12-Eve3 + May-13 SRD-421 combination is structural at $|\mu_B| \geq 8$, but the precise lower threshold of $q$ for the $(4, 3, 1^q)$ application should be cross-checked against the actual chain of inputs.
- Verifying Remark **??** explicitly on a multi-corner r=3 shape (e.g. $(5, 3, 2, 1^*)$ or $(5, 4, 1^*)$). The closed-form $\dim B = f^{\lambda^{(\geq 2)}}$ from the May-13 SYT-formula paper should now be derivable from r=2 (or r=3) additivity plus the inner chain reduction.

**Connection to the closed-form SYT theorem.** The May-13 `2026-05-13-closed-form-syt.tex` paper conjectured $\dim B^{(\lambda)} = f^{\lambda^{(\geq 2)}}$ in the stable regime. Theorem **??** (and its r=3+ generalisation in Remark **??**) suggests an inductive proof: $f^{\lambda^{(\geq 2)}}$ satisfies the same recursion $\dim B^{(\lambda)} = \sum_i \dim B^{(\mu_i)}$ by SYT branching, and the r=2 additivity matches it at the base. A full structural proof of the closed-form SYT theorem may now be within reach, provided the r=$r$ additivity holds in its claimed generality for all shapes in the stable regime.